

BACHELOR'S PROJECT

Research Proposal - Reinder Kas, s3189953

1 Background

1.1 Theoretical/Practical background

This project assumes general knowledge of Constraint Satisfaction Problems (CSP) as it will focus on a specific field within the space of CSPs.

In a world where mass production has become the norm, it is useful to think of ways to speed up the process time for a customer ordering and receiving a (complex) product. One way to realize this is to think of a product as a configuration for which all logic can be gathered into a single puzzle. Such a puzzle is called a Feature Model and an example of a simple phone Feature Model is shown in figure 1.1.

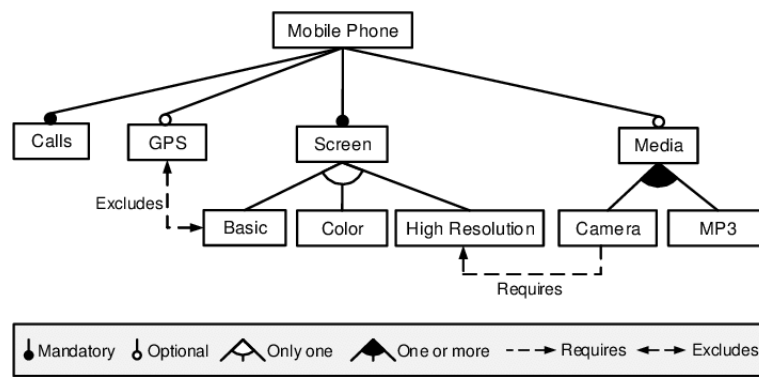


Figure 1.1: Visual representation showing a Feature Model of a simple phone.

These Feature Models exist of pre-defined rules (constraints) which determine what is and what is not possible within a product. Being that a Feature Model is a puzzle that can be built in any way that the pre-defined constraints allow, it is possible that a Feature Model is created for which there is no solution (e.g., two mandatory variables that exclude each other). With smaller Feature Models it is possible to diagnose by hand what the flaw in the model is. The problem is that as models get larger and more complex, this process gets more and more difficult. For large and complex models, the best way to do this is to use tools like an inference engine or a proof generator (e.g., z3). The problem with these options is that there can be difficulty in translating the specific business logic of the Feature Models into representations that can be used by such tools.

1.2 Previous literature

What is already known?

Plenty of research has already been done into Feature Models and the methods to diagnose problems within them. Books have been written about how to tackle these issues. What is generally accepted within Feature Models is that there are three categories of Constraints:

- **Non-Cross-tree Constraints:** These constraints form the basic structure of the Model and define the hierarchy within it (Mandatory/Optional/Single/Multiple). It is impossible to build a model with solely these constraints for which there is no solution.

- **Cross-tree Constraints:** These constraints (Exclude/Requires) are the constraints that define the relationship between the hierarchical structured Variables. It is generally accepted that only these types of constraints create problems in a Feature Model, as they do not define hierarchy but relationships within this hierarchy. This means that these are the constraints that cause a Feature Model to not have a solution (i.e., be unsatisfiable).
- **User Requirements:** Specific Variables within a Feature Model that the customer explicitly states to be present (i.e., their value must be ‘true’). A Feature Model can have a general solution if there are no User Requirements but could be unsatisfiable if there is a specific set of User Requirements present.

What questions/issues remain?

The issue with the current research is that it is based on very simple pre-defined Constraints. More specific implementations are necessary as the products get more and more complex. This increase in complexity means that the current state-of-the-art research findings are not always the optimal ones. One of these sub-optimal solutions is to remove a constraint from the Feature Model, which is a good solution if it solely involves Excludes and Requires. The problem here is that currently used implementations rarely exist of only Excludes and Requires Non-Cross-Tree constraints. For these more complex models, a new method of diagnosing problems within should be found (e.g., either by inference engine or proof generator).

1.3 Research Question

How can the minimal cause for unsatisfiability in Feature Model CSPs be detected and resolved efficiently?

2 Locate your project on the 2 dimensions

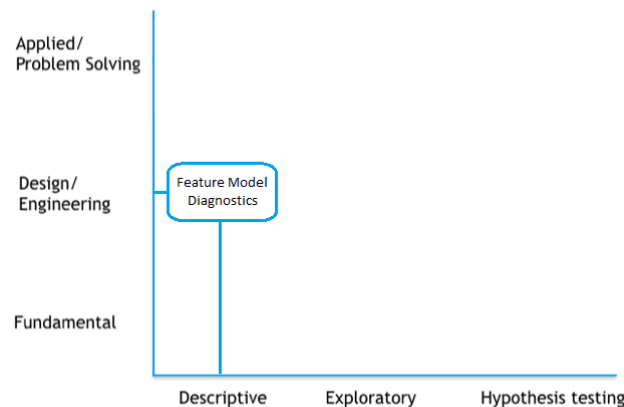


Figure 2.1: The type of research of my project with respect to the two dimensions discussed during the tutorial.

I think it is best to describe my project as a Descriptive project for now, as there will be no hypothesis testing within it. Furthermore, I think it is best to describe it as being a Design/Engineering type of project. This is because I will be looking at the possibility of finding a solution for a very specific implemented version of a Feature Model definition. It will be difficult to compare different methods of finding a solution for my problem as I suspect that it will be difficult to find a single solution in the first place.